

Task 2 Report

Synthesis & Gate-Level Simulation of a 4-Bit Multiplier Using Verilog HDL

InternSpark – VLSI Internship

Objective

The objective of this task is to design a 4-Bit Multiplier using Verilog HDL, perform behavioral simulation, synthesize the RTL design using Xilinx Vivado, and analyze the synthesized hardware implementation.

Software Used

- * Xilinx Vivado 2025.2
- * Verilog HDL
- * XSim Simulator

Project Description

A 4-Bit Multiplier multiplies two 4-bit binary inputs and produces an 8-bit output. The multiplier is implemented using the multiplication (*) operator in Verilog HDL. Vivado synthesizes this RTL description into digital hardware consisting of combinational logic.

Design Methodology

1. Created a new Vivado RTL project.
2. Developed the Verilog module for the 4-Bit Multiplier.
3. Designed a testbench with multiple test cases.
4. Performed RTL behavioral simulation.

5. Ran logic synthesis.
6. Generated the synthesized schematic.
7. Analyzed the utilization report.

Simulation Results

Several input combinations were tested to verify the correctness of the multiplier.

Input A		Input B	Product
2	3	6	
5	4	20	
7	6	42	
8	3	24	
15	15	225	

The observed outputs matched the expected multiplication results.

Synthesis Results

After synthesis, Vivado generated:

- * Synthesized Netlist
- * Hardware Schematic
- * Resource Utilization Report

The synthesized design correctly implemented the multiplier using FPGA logic resources.

Observations

- * RTL simulation verified functional correctness.
- * Synthesis successfully converted RTL into hardware logic.
- * Resource utilization information provided insight into FPGA implementation.

* The synthesized design matched the intended functionality.

Conclusion

The 4-Bit Multiplier was successfully designed, simulated, and synthesized using Verilog HDL in Xilinx Vivado. Both behavioral simulation and synthesis confirmed correct operation, providing practical experience with the RTL-to-hardware design flow used in digital VLSI systems.